Homework 3-SSIE 553

Neha Patankar
Spring 2026
Due-02/23/2026 by 11:59pm through Brightspace

Question 1[30pts]: Consider the following LP (*Same Problem 2 of HW2*):

$$\begin{aligned}
&\text{Min} \quad z = -2x_1 + x_2 - x_3 \\
&\textit{Subject to:} \\
&x_1 - x_2 + 3x_3 \leq 4 \\
&2x_1 + x_2 \leq 10 \\
&x_1 - x_2 - x_3 \leq 7 \\
&x_1 \geq 0, x_2 \geq 0, x_3 \geq 0
\end{aligned}$$

For each iteration k of the simplex method determine $x_{B^k}, x_{N^k}, d^k, B^k, N^k, \lambda^k$. *Note: For x_{B^k}, write the vector both in terms of values and variables while for x_{N^k} just the vector of variables will be fine.*

Question 2[20+5pts]: Consider the following Linear Programming Problem:

$$\begin{aligned}
&\text{Max} \quad z = x_1 + x_2 \\
&\textit{Subject to:} \\
&x_1 + x_2 + x_3 \leq 1 \\
&x_1 + 2x_3 \leq \frac{1}{2} \\
&x_1 \geq 0, x_2 \geq 0, x_3 \geq 0
\end{aligned}$$

i Solve the LP using the Simplex Method.

ii Does this problem has Alternative Optimal Solution? If Yes, determine another extreme point that is optimal and if No, Why?

Question 3[5+20pts]: Consider the following Linear Programming Problem:

$$\begin{aligned}
&\text{Max} \quad z = 2x_2 \\
&\textit{Subject to:} \\
&x_1 - x_2 \leq 4 \\
&-x_1 + x_2 \leq 1 \\
&x_1 \geq 0, x_2 \geq 0
\end{aligned}$$

i State the rule that can the used to determine if this LP is unbounded.

ii Use the rule stated in (i) to show that is LP is unbounded.

Question 4[10+5+5pts]: Consider the following Linear Programming Problem:

$$
\begin{aligned}
&\text{Max} \quad z = 5x_1 + 3x_2 \\
&\textit{Subject to:} \\
&4x_1 + 2x_2 \leq 12 \\
&4x_1 + x_2 \leq 10 \\
&x_1 + x_2 \leq 4 \\
&x_1 \geq 0, x_2 \geq 0
\end{aligned}
$$

i Solve this LP using the Simplex Method till we get a Basic Feasible Solution (BFS) of this LP that will confirm that this LP is degenerate. Lets call this BFS a *degenerate BFS*

ii Graph the feasible region of this LP to determine the constraints binding at the *degenerate BFS* found in part (ii).
(Hint: Please don't forget that when we study BFS, we only consider the LP in the Standard Form)

iii Are number of constraints binding at the degenerate BFS greater than n?
(n is the total number of variables in $Ax = b, x \geq 0$)